U.S.T.H.B. / Faculty of Mathematics
Department of Operations Research

Academic year: 2026/2027
Subject: Thor (BigDataA, S1)

Exercise Sheet no. 1 (General Concepts)

Exercise no. 1 :

a) A dentist sees n patients arrive when his office opens. Assuming he knows in advance how much time he will spend with each patient, which criterion should he consider in order to minimize the average waiting time of these n patients?

b) Now suppose that the clients do not arrive at the same time, but that he knows each client's arrival date in advance. Which criterion should he consider in order to minimize the average waiting time of these n patients?

Exercise no. 2 : Show that the criterion C_{max} is a special case of the criterion L_{max}. Likewise, that the criterion $\overline{C}$ is a special case of the criterion $\overline{L}$.

Exercise no. 3 : Show that $\overline{C_w}$, $\overline{F_w}$ and $\overline{L_w}$ are equivalent, and in particular $\overline{C}$, $\overline{F}$ and $\overline{L}$.

Exercise no. 4 : Consider a scheduling problem made up of one machine and two tasks with zero release dates, and with the processing times and due dates given below.

T_i	T_1	T_2
p_i	3	1
d_i	3	4

Compute $\overline{L}$ and $\overline{T}$ for the two possible schedules, and deduce that a schedule that is optimal for the criterion $\overline{L}$ is not optimal for the criterion $\overline{T}$, and vice versa.

Exercise no. 5 : Show that a schedule that is optimal for L_{max} is also optimal for T_{max}.

Exercise no. 6 : Give an example of a scheduling problem where an optimal solution for T_{max} is not optimal for L_{max} (consider only two tasks and a single machine).

Exercise no. 7 : A maintenance workshop of a construction company is made up of 5 platforms, each able to hold a single vehicle. To avoid breakdowns, each of the company's vehicles goes to the workshop, after a certain period in service, for an inspection. The duration of the inspection depends on the vehicle and on the type of inspection to be carried out. We want to define an inspection schedule that minimizes the average waiting time of the vehicles.

Classify this scheduling problem.

Exercise no. 8 : Classify the four problems seen in class.

Exercise no. 9 : A company carries out a project made up of 6 tasks using machines. For each of these tasks, the duration and the due date are known (see the table below).

Task	A	B	C	D	E	F
Duration (in days)	7	7	3	6	2	5
Due date	15	15	9	10	5	10

Give a schedule, optimal if possible, and the corresponding value of the objective function for each of the following problems:

a) $1||C_{max}$.
b) $1||\Sigma C_i$.
c) $1||L_{max}$.
d) $1||T_{max}$.
e) $1||\Sigma u_i$.
f) $P2||C_{max}$.
g) $P2||\Sigma C_i$.
h) $P2||L_{max}$.
i) $P2||T_{max}$.
j) $P2||\Sigma u_i$.

Exercise no. 10 : A company carries out a project made up of 6 tasks using two specialized machines. For each of these tasks, the duration on each of the two machines is known (see the table below).

Task	A	B	C	D	E	F
Duration on M_1	5	5	1	4	1	3
Duration on M_2	3	1	4	5	3	4

Give a schedule, optimal if possible, and the corresponding value of the objective function for each of the following problems:

a) $F2||C_{max}$.
b) $F2||\Sigma C_i$.
c) $O2||C_{max}$.
d) $O2||\Sigma C_i$.

Exercise no. 11 : Describe, using a diagram, the polynomial reductions (complexity hierarchy) between the following scheduling problems:

a) $1//\overline{C_w}$
b) $1/d_i = d/\overline{T_w}$
c) $1/p_i = 1/\overline{T_w}$
d) $1//\overline{T_w}$
e) $Pm/p_i = 1/\overline{T_w}$
f) $Pm//\overline{T_w}$